

Aaditya Ghosalkar

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Education

University of Virginia, School of Engineering & Applied Sciences

Charlottesville, VA

B.S., Computer Science

2021 - 2025

Relevant Courses: Data Structures & Algorithms, Computer Systems & Organization, Artificial Intelligence, Machine Learning, Data Science, Probability & Statistics, Network Security

Certifications: AWS Certified Solutions Architect (Associate), AWS Certified AI Practitioner, PCAP

Skills

Programming Languages: Python, JavaScript, C++

AI & ML: Generative AI, LLMs (LLaMA-2, OpenAI API), NLP, Prompt Engineering, pandas, MCPI, Langchain, LangGraph

Frontend: React.js, Angular, HTML5, Responsive UI Design

Backend: Django, RESTful APIs, Flask, FastAPI, Microservices, Authentication and RBAC, Springboot

Databases: PostgreSQL, MySQL, NoSQL

Cloud & DevOps: AWS, Docker, Kubernetes, CI/CD

Projects

Implicit Bias & Toxicity Detection Model | Python, Machine Learning, NLP

- Trained and evaluated an NLP classification model on the ToxiGen dataset (~27,000 samples) to detect implicit bias and subtle toxicity in text, achieving a macro F1 score of 0.74 — a ~12% improvement over the zero-shot baseline, going beyond surface-level profanity filtering.
- Experimented with LoRA-based fine-tuning and feature engineering techniques to improve detection of nuanced, non-explicit harmful language; reduced false-negative rate on implicit bias examples by ~18% compared to the base model.

AI Jewelry Image Generation App | Python, AWS, Generative AI

- Designed and deployed a client-facing jewelry image generation application from scratch, owning the full development lifecycle from initial requirements to cloud deployment on AWS.
- Designed a scalable Python backend with RESTful API endpoints, containerized for consistent deployment; system maintained 99.5% uptime over the first month of client use.

Professional Experience

Full Stack Software Developer, Hilex Technologies LLC, Herndon, VA

November 2024 - Present

- Developed and deployed production ready web applications using React front end and Python backend services, owning the full development lifecycle – from requirements gathering to deployment.
- Built and maintained CI/CD pipelines using GitHub Actions, deployed containerized services to Azure Kubernetes Service and GCP environments.
- Designed and deployed policy-aware text generation services, exposed via RESTful API, enabling secure and scalable AI features.
- Built document parsing and ETL pipelines to process and transform structured/unstructured data for downstream services.
- Collaborated with clients to design and deliver proof-of-concept applications, translating requirements into functional prototypes.
- Applied data processing and transformation techniques to prepare structured and unstructured datasets for AI model consumption.

Software Engineer Intern, Avalon Solutions, New York, NY

November 2023 – April 2024

- Developed and maintained production grade Python based REST APIs handling 1,000+ daily requests, with structured logging, robust exception handling and optimized SQL stored procedures.
- Collaborated in Agile sprints across a team of 6, contributing to code reviews, unit testing, and integration testing for high-volume API services.
- Contributed to frontend components using React, HTML, CSS, and JavaScript to support API-driven features.